SIKORSKY UH-60M BLACKHAWK

FOR X-PLANE

Version 0.8.dX

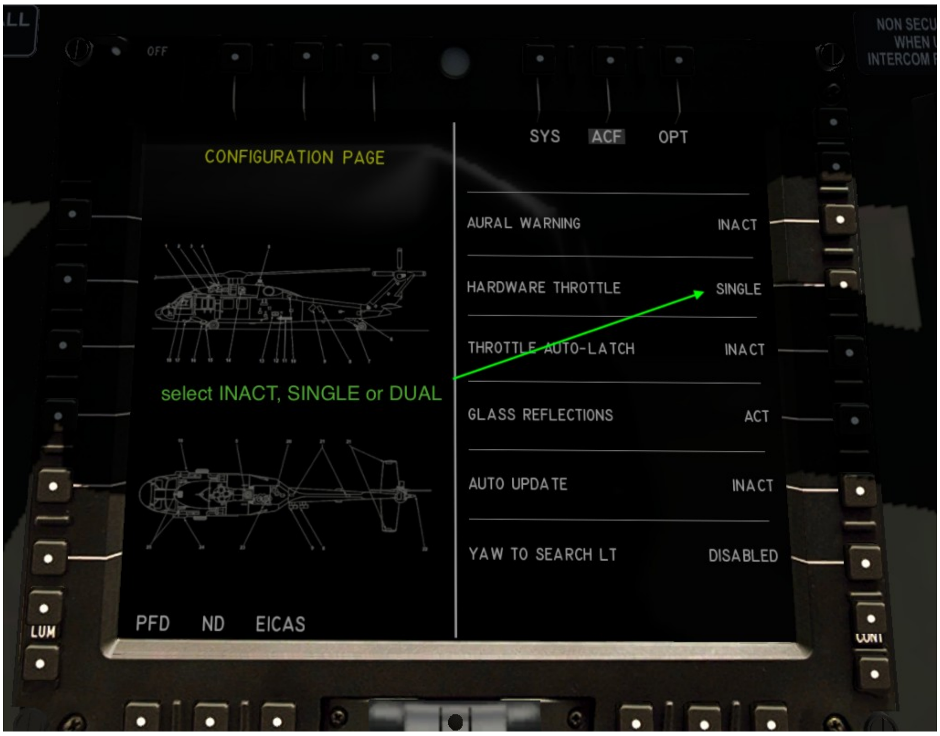
Overview

Quick Start

- Make sure you **do not** have your joystick mapped to the „throttle“ axis. If you have a hardware throttle, map your joystick axis to „throttle3“, and if you have a dual-hw-throttle, map the 2nd to „throttle4“ !

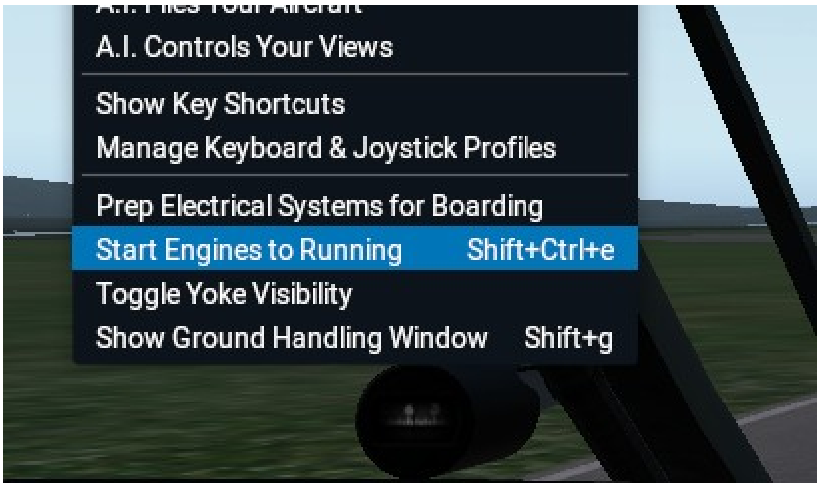
- In the EICAS → CONF → ACF tab, select which setup you have.

INACT will allow you to control the throttle levers in overhead panel by mouse or VR controller !



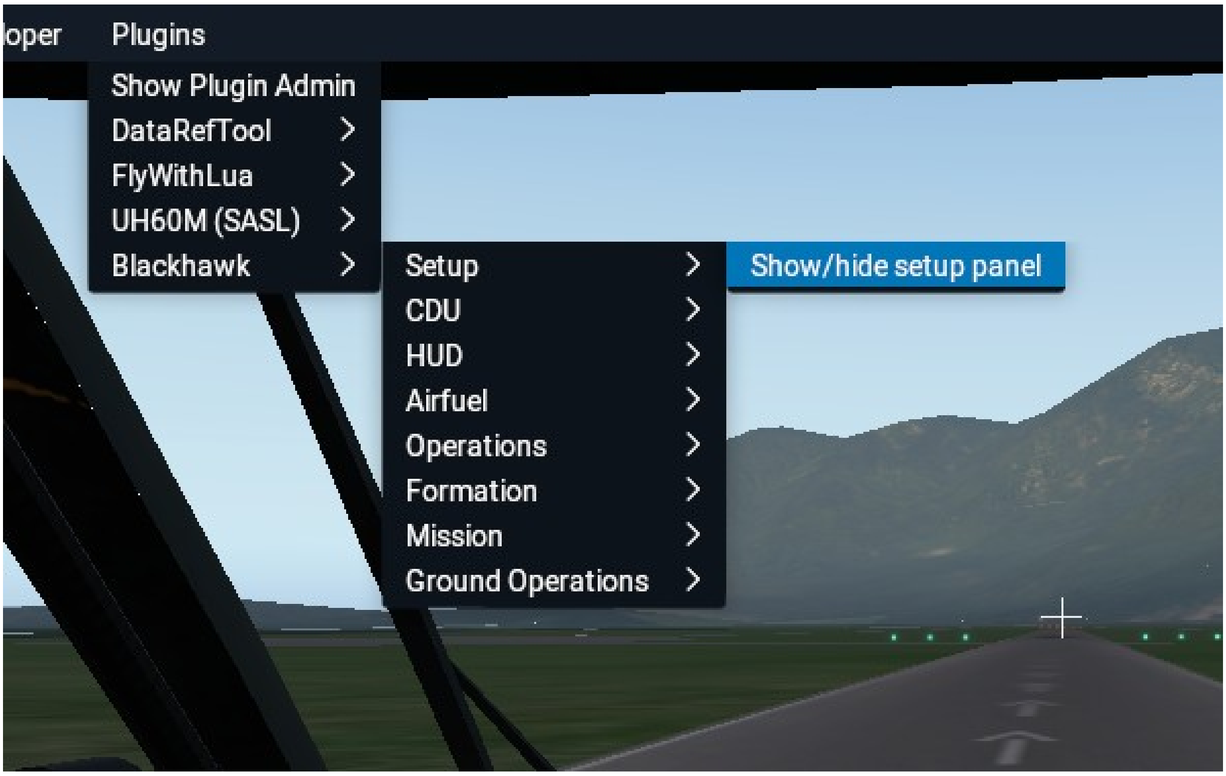
- Start the helicopter

either with „running engines“ or use the „autostart“ option from the „Flight“ pull-down menu.

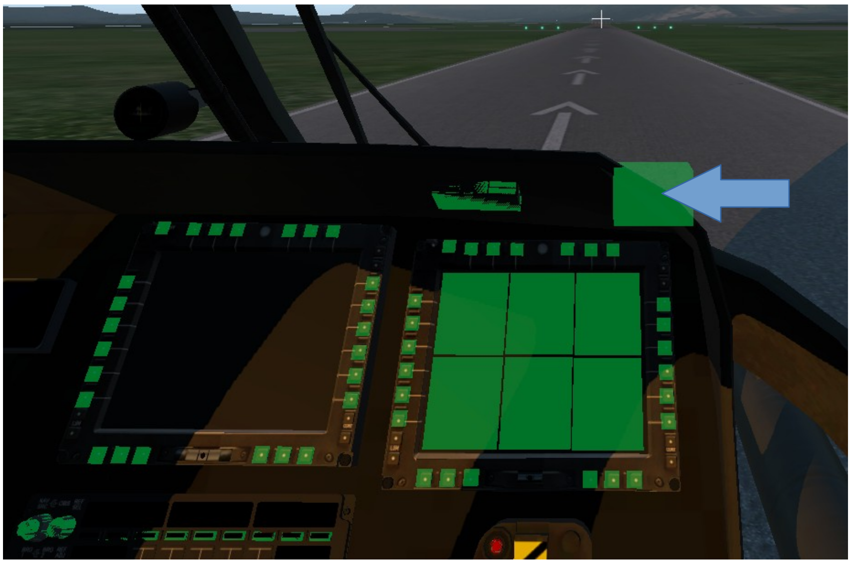


Configuration

To configure the Blackhawk you can open the „setup panel“ from the „Blackhawk“ setup sub-menu inside the „Plugins“ pull-down menu.



Alternatively you can click on the upper-right-corner of the cockpit dash. If uncertain use the „Show instrument click regions“ under the „View“ pull down menu.

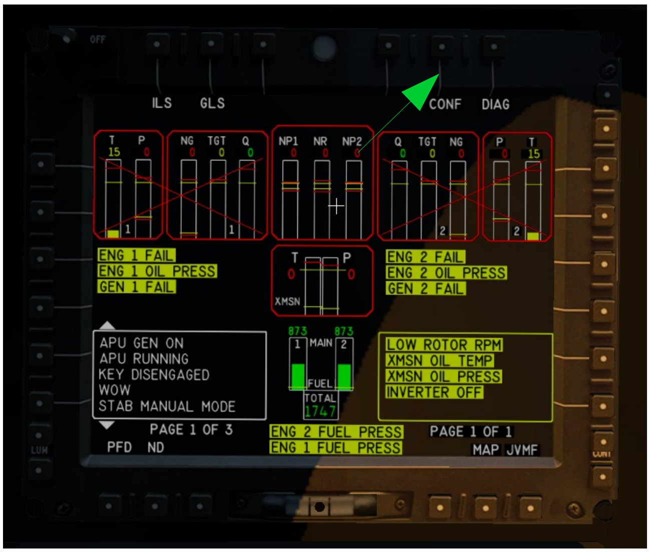


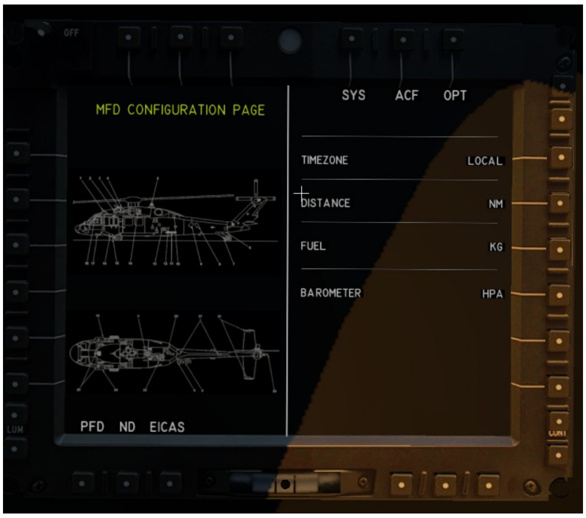
Config Menus


MAIN
CAMS
CONF
PREFS
GUNS
MISSILE SENSORS
MISSILES & ROCKETS
CHAFF & FLARES
RADAR
X FLIR CAM
X HOIST / WINCH
HRST / FAST-ROPE
FUEL PROBE
IRCM / IR JAMMER
X ESSS PYLON
2 ESS AUX TANKS
1 SEAT ROWS
3 PASSENGERS
UNITED STATES ARMY

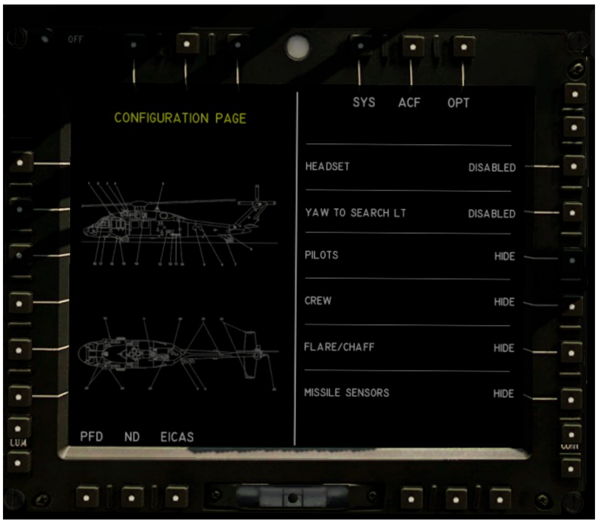

MAIN
CAMS
CONF
PREFS
X SHOW PILOTS
SHOW CREW
GLASS REFLECTION
X ROTOR DOWNWASH
X SHOW CKPIT DOORS
X HEADSET
X ARMOR PANEL
X VIBRATIONS
0.06 FPS DEAD ZONE
B CYLC RE-CENTER MODE
450

Some system default settings can be set in the Multi-Function-Displays (MFD) under the „CONF“ button on the „EICAS“ screen (of course batteries or APU have to be switched ON first).





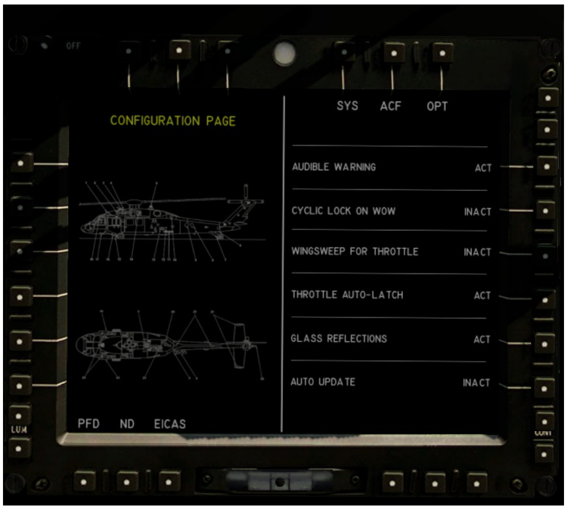
showing the CONF SYS



ACF

and

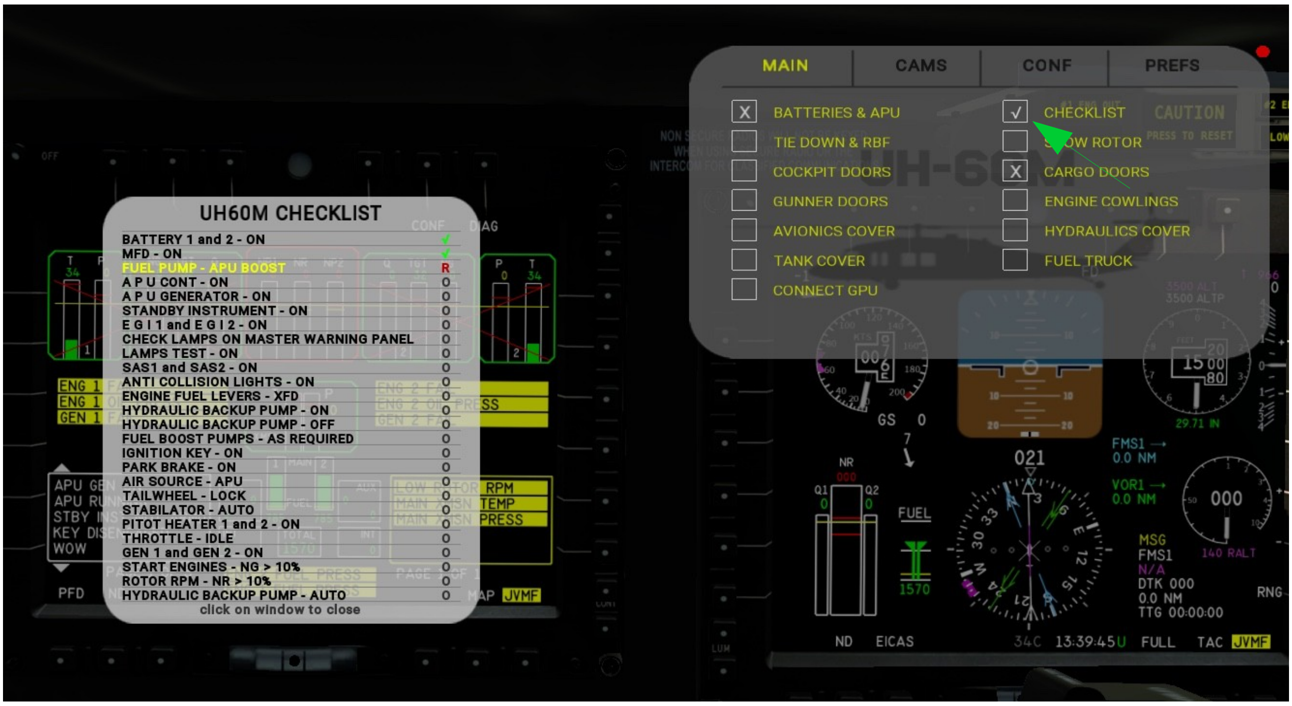
OPT tab



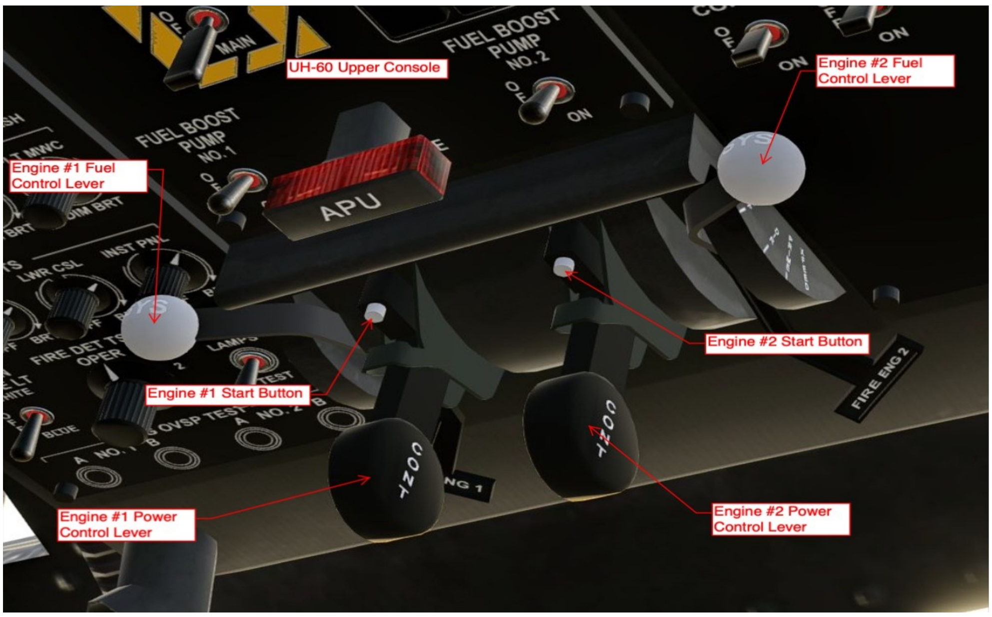
Note: Settings and configurations are saved in either the acf_config.ini located in the acf folder or in the opt_config.ini located in the selected livery folder, depending whether they are „global“ or „livery“ related.

Cold & Dark

On the SETUP panel you can select the CHECKLIST option. A popup will guide you interactively through the steps required to start the helicopter manually from cold&dark.



At the end of the list the engine-start can either be performed by pressing the white buttons on the throttle levers located on the over-head panel, or by assigning the appropriate command to a joystick button (see “Datarefs and Commands” section below).



Please note that the "Ignition Key" needs to be set to the "ON" position !

Check messages on the EICAS screen and confirm them by pressing the CAUTION button or via the mapped command. Audible alarm can be configured (ON/OFF) in the CONF menu on the MFD.

CDU – Control and Display Unit
FMS – Flight Management System

The CDU or FMS is being used to control communication and navigation systems, flight plans and missions. A mission is similar to a flight plan but has additional functionality such as runnings X-Plane commands depending on programmable conditions. More about it in the “Missions” section.


CDU1
RNG 0.0 GPS 13:03:53
BRG 000 TTG 00:00:00
NXT 003 MSL 1784 GS 0
WND 0 / 0 K--- LOWI
FPN WPTS
ETSI Ingolstadt
RUN
ANKER waypoint (3000/300)
EDDM Munich (4000/200)
FUEL
LATLO waypoint (5900/400)
MAP
VAROB waypoint (6500/300)
RTN


CDU1
RNG 0.0 GPS 13:04:12
BRG 000 TTG 00:00:00
NXT 003 MSL 1784 GS 0
WND 0 / 0 K--- LOWI
FPN EDIT
ETSI (48.72/11.53) Ingolstadt
ADD
ANKER (48.57/12.00) waypoint (3000/300)
INS
EDDM (48.36/11.79) Munich (4000/200)
DEL
LATLO (47.68/12.81) waypoint (5900/400)
ALT
VAROB (47.63/12.54) waypoint (6500/300)
RTN


CDU1
RNG 0.0 GPS 13:04:30
BRG 000 TTG 00:00:00
NXT 003 MSL 1784 GS 0
WND 0 / 0 K--- LOWI
GPS
EDPK , 0km/0nm Schönberg
DRCT
EDPC , 20km/11nm Bad Endorf
XEDAF , 21km/11nm Altötting
EDJG , 21km/11nm Grabenstätt
EDNV , 24km/13nm Vogtareuth
RTN


CDU1
RNG 0.0 GPS 13:04:51
BRG 000 TTG 00:00:00
NXT 003 MSL 1784 GS 0
WND 0 / 0 K--- LOWI
PRESETS
COM
NAV
GPS
RTN


CDU1
RNG 0.0 GPS 13:05:53
BRG 000 TTG 00:00:00
NXT 003 MSL 1784 GS 0
WND 0 / 0 K--- LOWI
NAV
NAV1 108.00
NAV2 108.30
ADF1 400
ADF2 115
PRESET
RTN
FREQ:[____108.00]


CDU1
RNG 0.0 GPS 13:05:27
BRG 000 TTG 00:00:00
NXT 003 MSL 1784 GS 0
WND 0 / 0 K--- LOWI
NAV PRE
02 THE-BALL 108.10
ILS-LOWI 07 111.10
03 TACAN-MAH 99X
ADF-MNW 08 338
04 VOR-MAH 115.20
ADF-MSW 09 400
05 VOR-MUN 112.30
06 ILS-ETSI 108.10
RTN


CDU1
RNG 0.0 GPS 13:06:08
BRG 000 TTG 00:00:00
NXT 003 MSL 1784 GS 0
WND 0 / 0 K--- LOWI
ZERO
MSN
ALL
COM
NAV
GPS
RTN


CDU1
RNG 0.0 GPS 13:06:26
BRG 000 TTG 00:00:00
NXT 003 MSL 1784 GS 0
WND 0 / 0 K--- LOWI
FUEL CALC
DISTANCE 397.3 KM
FUEL FLOW 0.0 LB/H
REQ FUEL N/A
FUEL FOR N/A
REM FUEL 3967 LBS
REM TIME N/A
KTS ACT
ETA
RTN

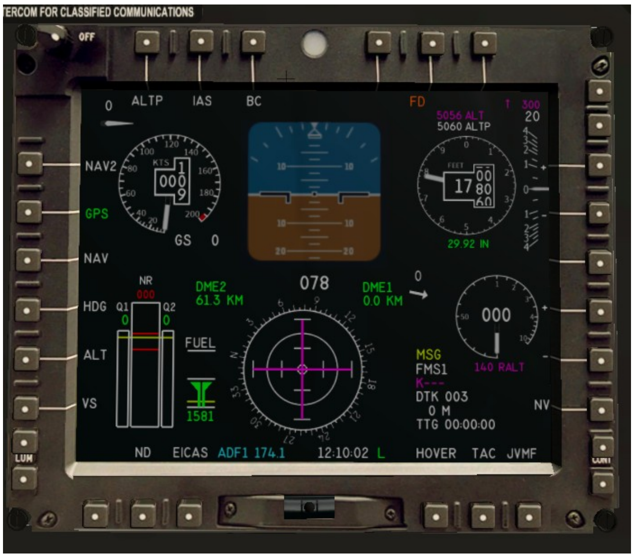
Clicking on the CDU displays in the pedestal will open a CDU popup. This can be helpful in VR. The popups are movable and resizable.

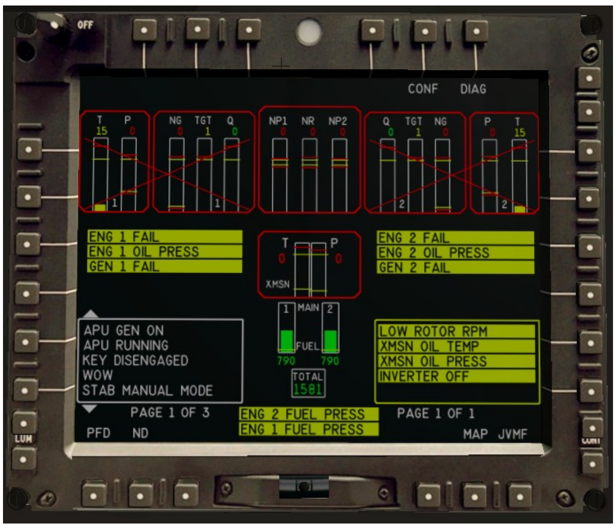
Read more in the UH60M-FMS manual located in the docs folder.

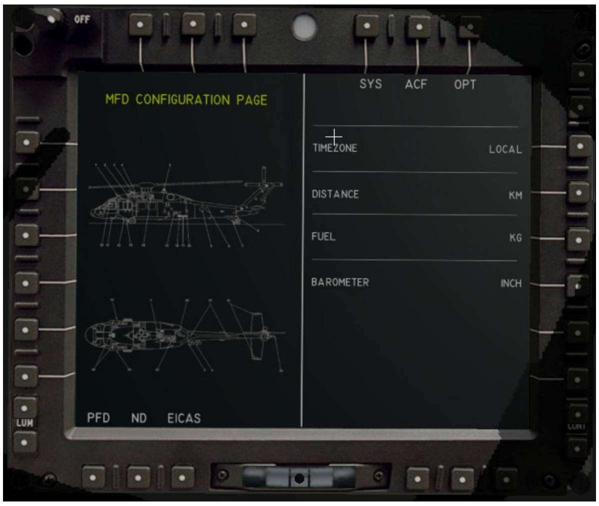
Click here to see an overview video of the CDU/FMS:

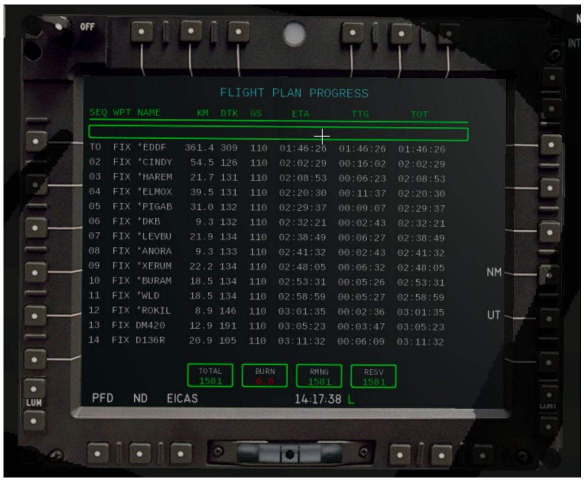
https://www.youtube.com/watch?v=gSlrfdNKChs

<u>MFD – Multi Function Displays</u>









Avitab support (also on iPad)

Camera guided target locking

JVMF MESSAGES

JVMF stands for Joined Variable Message Format.

It is kind of a military email system. In the simulation it is used to create "tasks" for the crew. Via the "Operations" pull down menu a task to extract a Seal from a random location can be created.

An OSM query in the background tries to find an appropriate spot in the area 20km around the current position. Once the message is being accepted (ACK = acknowledged), the FMS is being programmed with the coordinates and so can lead the pilot to the Seal.

The JVMF-softkey switches to YELLOW as soon as a new "unread" message shows up in the Inbox. The symbols "greater-than" and "?" indicate "un-read" and "un-acknowledged" messages.

See: https://www.youtube.com/watch?v=2KXv8P6lc_w

FDCP – Flight Director Control Panel

The FDCP is the control panel for the autopilot.

The 5 rotaries for RALT (radio altimeter) , ALTP (programmed altitude). ALT (altitude hold), IAS (indicated airspeed) and HDG (heading) can be used to change the values in the displays. Pushing the rotaries syncs the display with the actual values (i.e. current altitude).

The black buttons under the displays allow to activate/deactivate the related "upper modes" AP (autopilot) function. If a function is activated it will be indicated by a greed LED inside the button. Some function also have an "armed" state, which is indicated by an amber LED.

The very right button in the upper row controls and show whether the FD (flight director) is engaged/coupled. If it is coupled (CPLD) then the FD controls the aircraft, otherwise the display shows DCPL which stands for de-coupled.

The very left button in the upper row engages RNAV (radio navigation) which means the selected navigation source is providing input to the AP. Valid sources are NAV1, NAV2, LOC1, LOC2 , FMS. They can be selected with the outer ring of the very left rotary in the upper row.

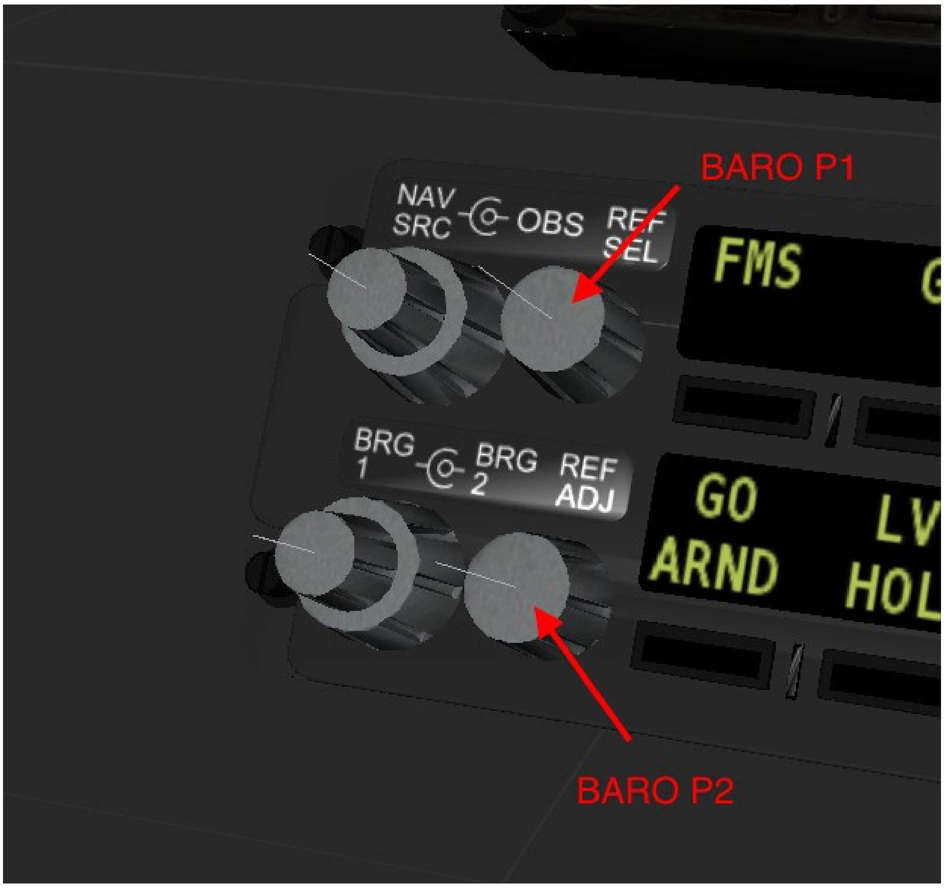
"GO ARND" = Go-Around will set the AP to climb with 700 ft/sec and 70 kts.

"LVL HOLD" = LevelHold enables the PID controller which keeps the current altitude-above-ground. It is armed (amber LED) unless vertical speed is less 0.1 m/sec and kias <= 50 kts. Then it will change to green.

"HVR POS" = HoverHold enables the PID pid controller which keeps the current position if the helicopter hovers with less the 1.5 m/sec horizontal speed.

NOTE: All AP functions are DISENGAGED if we have WOW (weight-on-wheels) or <= 50 kias. Also all weapons will be disarmed with WOW !!

<u>The pilots and co-pilots barometric pressure can be set using the 2 knobs on the left side of the FDCP</u>



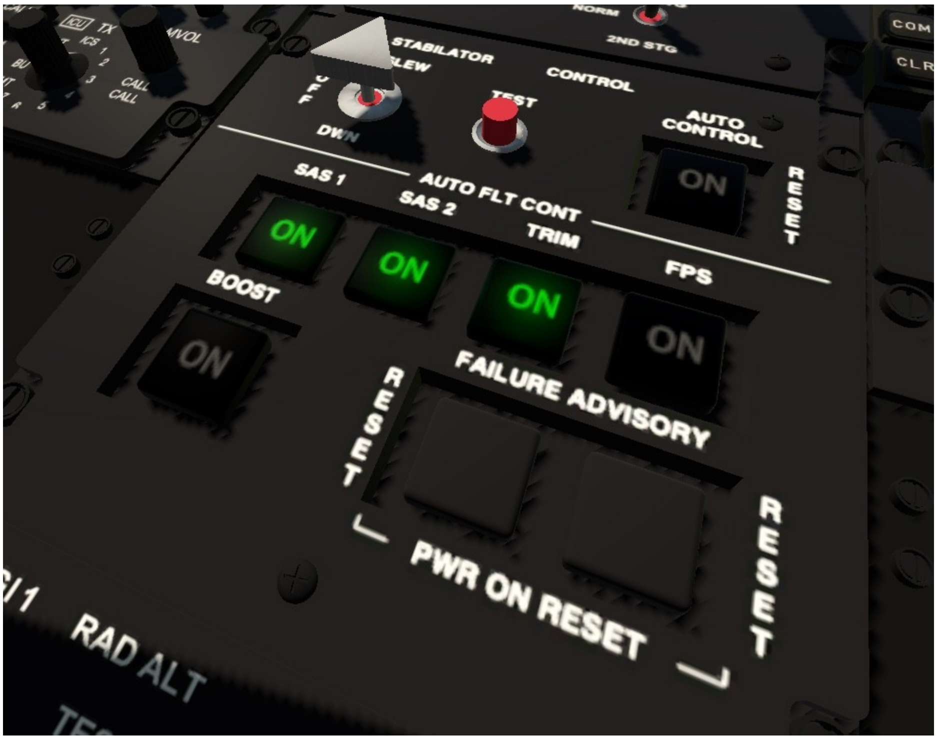
Changing between Inch and hPa can be made in the MFD config page under the SYS tab.

Read more in the UH60M-FDCP manual located in the docs folder.


6042
U.S.COAST
GUARD
A5
S→


6041
U.S.COAST
GUARD

Flight Control Functions



SAS1 / SAS2 is a redundant Stability Augmentation System. Each controller can be used isolated (with reduced stability) or joined together.

SAS Boost increases the agility/responsiveness of the stability system.

TRIM (when switched on) allows to re-center the cyclic using FTR. There are 2 modes. Mode “A” behaves as before: press FTR, move the cyclic to the center and release FTR. Pressing FTR for more than 3 seconds will center the cyclic back to 0/0. Mode “B” will store the new center if FTR gets “released” and allows you to position the cyclic back to the center within 3 seconds. Both are useful for spring-loaded cyclic joysticks. TRIM also enables the beep-trim switches to trim the aircraft pitch and roll. The state of the switch is persistent.

FPS (flight path stability) provides a Pitch and Roll Attitude-Hold using the autopilot, if no upper-modes like VS, ALT, HDG or IAS are selected. To change the attitude, simply move the cyclic into the desired direction and then back to center. FTR will pause FPS. FPS will fall back to armed on WOW or if upper AP modes are engaged. The state of the switch is persistent.

BeepTrim functions can be used by mapping the following commands to the coolie-hat switch of your cyclic joystick.

uh60m/sys/beep_fwd	Beep trip forward
uh60m/sys/beep_rev	Beep trim backward
uh60m/sys/beep_lft	Beep trim left
uh60m/sys/beep_rgt	Beep trim right

These switches have different functions, which vary depending on the context they're used in.

1) If the autopilot is not active, the 4 switches control the aircraft trimming (aileron & elevator). However, if the "search light" is switched on, they control the search light direction and if the FLIR-Cam is powered on, they control the camera pitch and heading.

2) If the autopilot is active (coupled or engaged), but no upper modes are selected, they will control the attitude-hold for pitch and roll with +/- 2 degrees per second if FPS is ON !

3) If the upper mode HDG is engaged, the left & right switches will decrease or increase the autopilot-heading with +/- 10 degree per second with a dynamic speed-up up to 20 deg/sec.

4) If the upper mode VS is engaged, the fwd & rev switches will increase or decrease autopilot-vertical-speed by 100 fpm per second.

5) If the upper mode IAS is engaged, the fwd & rev switches will increase or decrease the autopilot airspeed by 10 kts/sec.

6) If the upper mode ALT is engaged, the fwd & rev switches will increase or decrease the ALTP altitude with 100 fpm/sec with a dynamic speedup up to 400 fpm/sec.

Force Trim Release

Command: uh60/sys/ftr

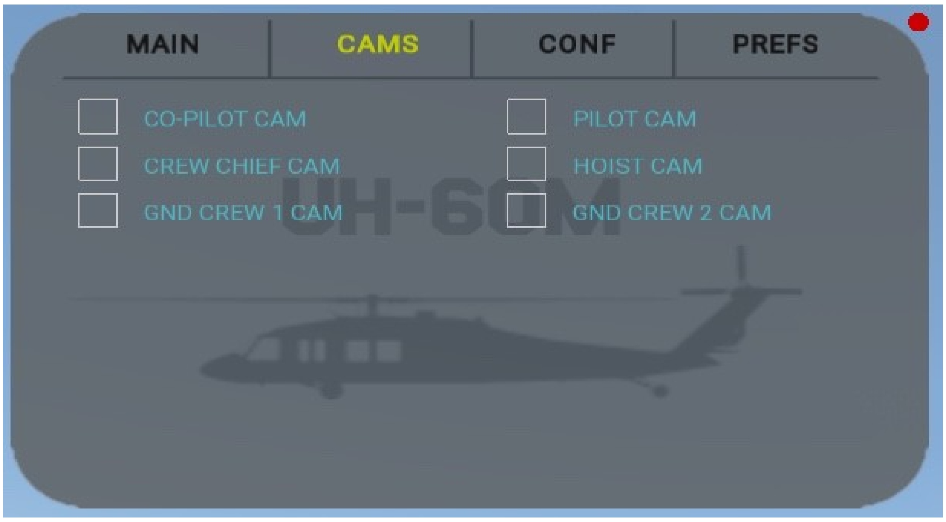
- With TRIM ON and AP OFF, FTR allows to re-center the cyclic. 2 modes can be chosen from the PREFS menu. In mode “A” you press FTR, move the joystick to a new center-position (cyclic in cockpit will NOT move while FTR is pressed!) and release FTR. In mode “B” you press FTR and move the joystick to a new position (cyclic in cockpit WILL MOVE while FTR is pressed!) and then release FTR. Now you can move your joystick back to center within 3 seconds.
  Holding FTR in mode “A” for longer than 3 seconds will slowly center it back to the 0/0 position.

- With AP-on, it will temporary disengage AP as long as the button is pressed. Also modes like level-hold, hover-hold will be disengaged and go-around will be paused.

- FTR will also reset the pitch and roll trimming and stop a SAR search pattern operation.

Cameras

The Blackhawk comes with a built-in multi-camera system. It allows you to do role-playing games.



The system tracks each individual cam position and also allows to save, load and recall position sequences (i.e. preflight check)

The cameras can be selected through the setup-menu or by using one of the available commands (see DataRefs and Commands).

For example in combination with LVL-HOLD and HVR-HOLD you can observe HRST operations as a Crew Chief while sitting in the Cargo Area.

Hint: make sure you engage your AP before you leave your pilots seat ;)

Each camera can also record positions which can be saved, restored and re-called afterwards. See preflight.cam in the “cams” folder. Check out the “uh60m/cam/” commands below.

Operations and Actions

HRST Fast-Rope-Extraction
Requirement: "HRST" option enabled in setup menu.
Command: uh60m/ops/drop_dude

Every time the command gets called, a Seal will slide down on one of the ropes and if on the ground, moves to a random "defense position". This also works on moving objects like the aircraft carrier.

Air-Refuel
Requirement: "REFUEL PROBE" enabled in setup menu.
Command: uh60m/ops/c130_show.

First command will call the C130. It will adopt to your heading and speed.
Second command will end operation. C130 will climb and disappear.
Third command will delete the C130 immediately.

Target tracking
Requirement: "FLIR CAM" enabled in setup menu.

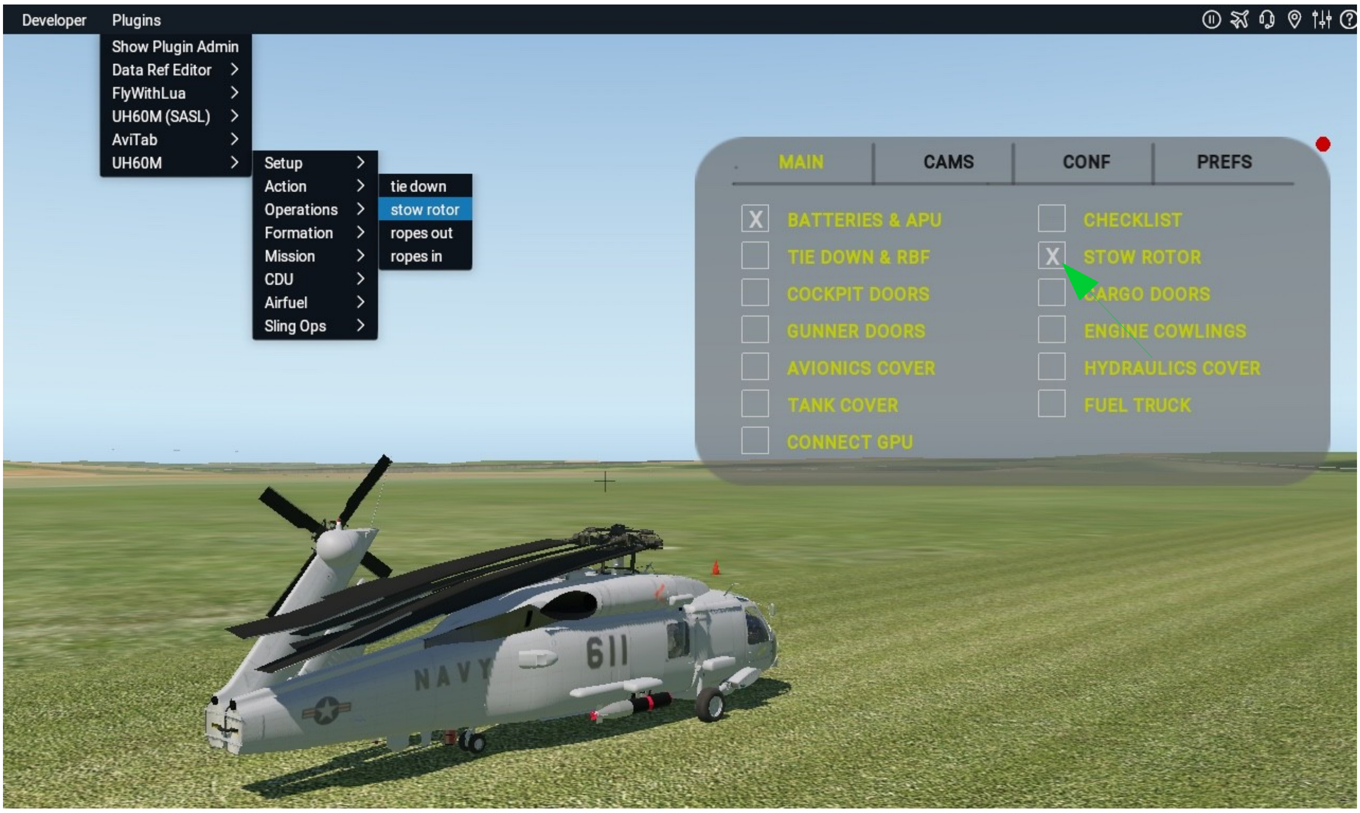
On MFD1 select TAC and then TGT. Clicking the mouse into MFD display will show a red aiming mark which allows to control the cam with the mouse. Pressing the LCK button (or use the equiv. command) will lock the camera to the aimed position. A yellow pole will show up at the locked location. A 2nd click will release the target and the camera moves back to the default position.

Seal Extraction
An OSM query is being used to select a random location in the range of 20km to place a Seal at. A JVMF order is being created to get him home. If the message is ACKnowledged, the FMS is programmed with the location. Fly to the Seal and land next to him to take him on board.

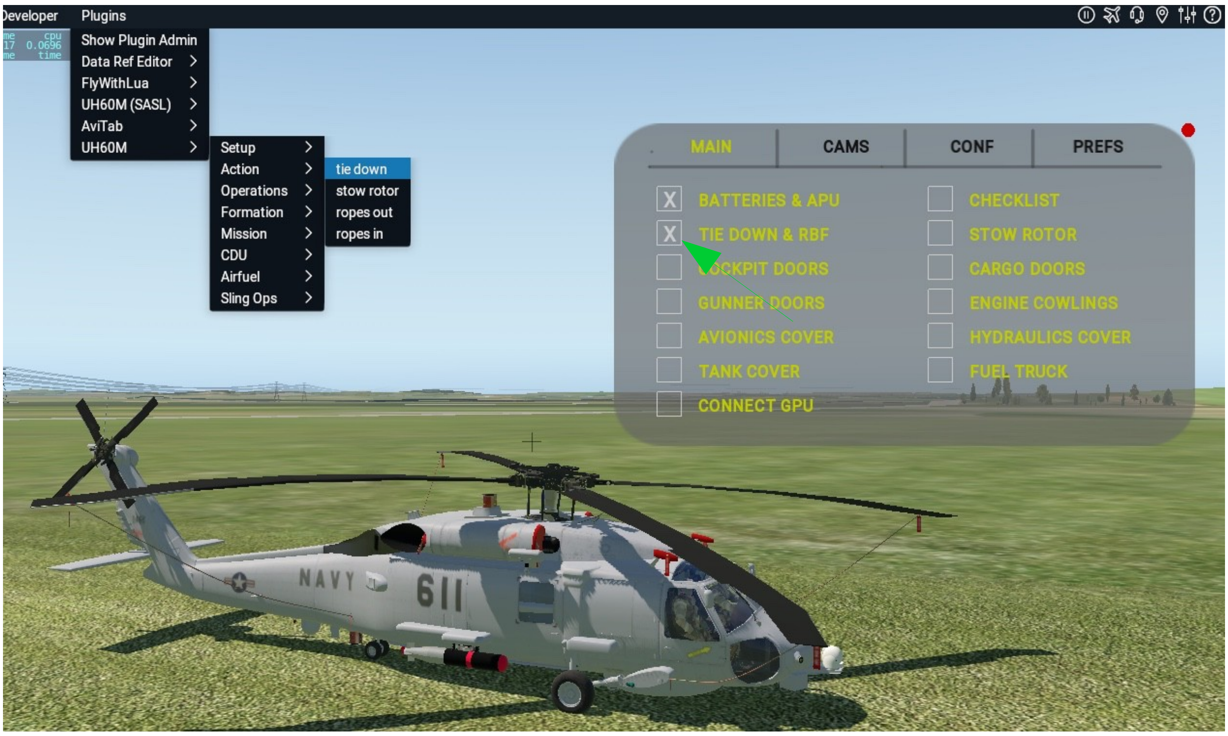
Seal Insertion
An OSM query is being used to select a random location in the range of 20km to insert a few Seals. A JVMF order is being created. If the message is ACKnowledged, the FMS is programmed with the location and the seals are loaded on board. Fly to the spot. Another Seal will play marshaller. Land next to him and let the seals exit the aircraft.

Note: might not work in areas where no valid OSM data is available.

Stow Rotor & Tail (fold/unfold)



Tie-Down and Remove-Before-Flight objects



Note: during tie-down and/or rotor-stow, ignition key is not inserted. So engine start is not possible !

Humvee and Howitzer Slung Ops

Place a Humvee or Howitzer randomly in the range of 100m using the pulldown menu “Operations/Slingops”. If you get closer to it (50m), it offers you a slung cable to connect. Hold an AGL of 15 ft and try to position the cargo hook above the cable. Once connected, the weight of the load gets added to the acf. After you lifted it above 20 ft AGL once, it will automatically disconnect if the object touches the ground again.

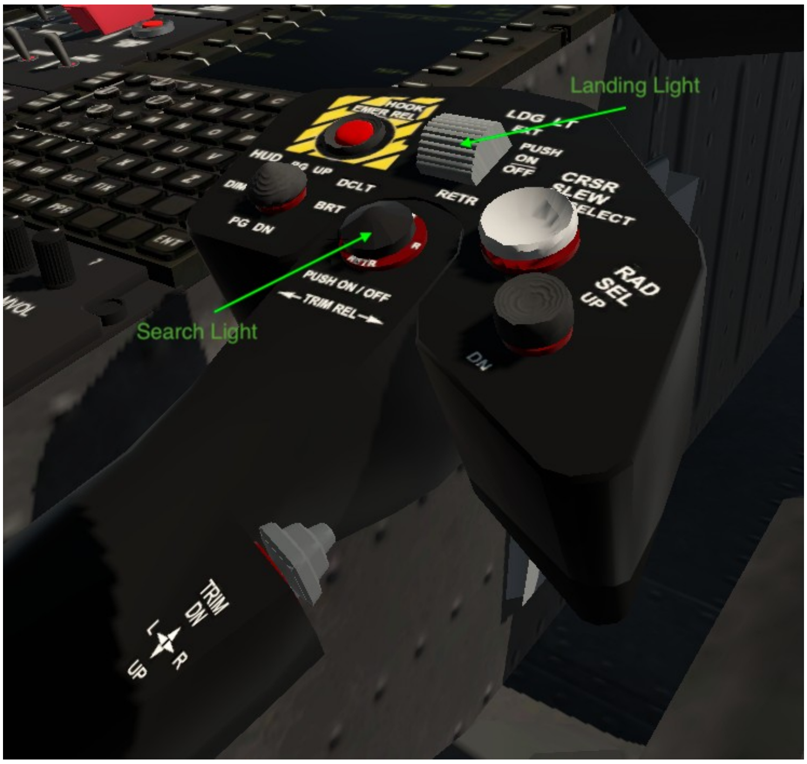
Lights

Search light

Use command "uh60m/lights/srch_lt_on" to switch search light on and off. Other commands to steer the search light: 'srch_lt_up', 'srch_lt_dn', 'srch_lt_lft' and 'srch_lt_rgt'. On ground (wow) the search light can be coupled to the yaw axis (pedals).

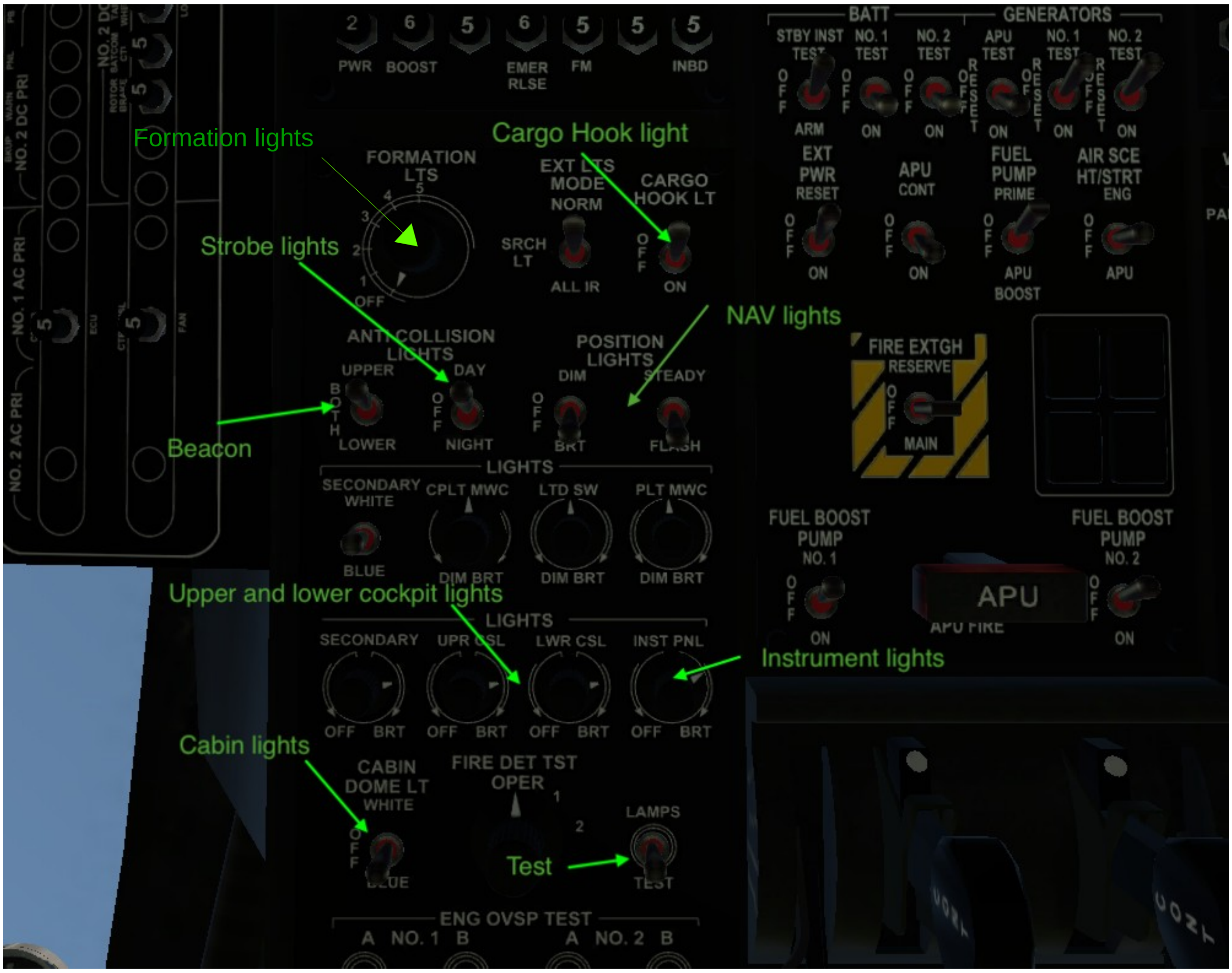
Landing light

Use command "uh60m/lights/ldg_lt_on" to switch landing light on and off.



Instrument lights

Cockpit, Cabin dome, Cargo Hook and external lights



Clock

Reversionary switch panel

The UH-60M has a number of redundant systems. The reversionary switch panel allows the pilot and co-pilot to switch between the primary and the secondary system. In the simulation these functions are limited by the available options provided by X-Plane.

At the moment it is possible to switch the ADC (air data) only. Depending of the switch setting, the MFDs show altitude (baro and radio), air speed, vertical speed and barometer setting from the "pilot" (green) or the "co-pilot" (amber) instruments.

Weapons

GUNS

Requirement: “GUNS” installed in setup menu
Both “gunner doors” need to be open !

The guns are tied to the yaw axis. So they will follow the input to the tail rotor pedals.

CHAFF

Command: uh60m/weapons/deploy_chaff

FLARE

Commands: uh60m/weapons/deploy_flares

MISSILE DETECTORS

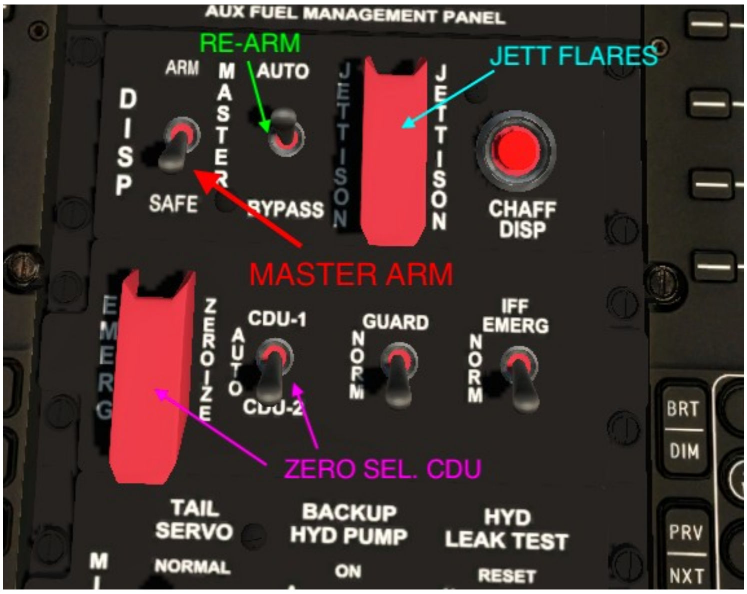
EOMS (electro-optic-missile-sensor) can be configured on the avionics cover and at the tail.

Note: weapons are disabled/armed as long as we have WOW (weight on wheels).

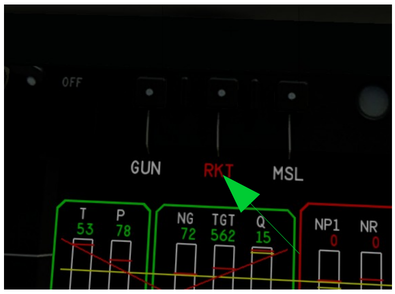
ARMING AND SELECTING WEAPONS

Use DataRef “uh60m/weapons/master_arm” or the master-arm-switch in the pedestal panel to arm/disarm all weapons together.

Use “auto-bypass-auto” to re-arm (re-load) your weapons.



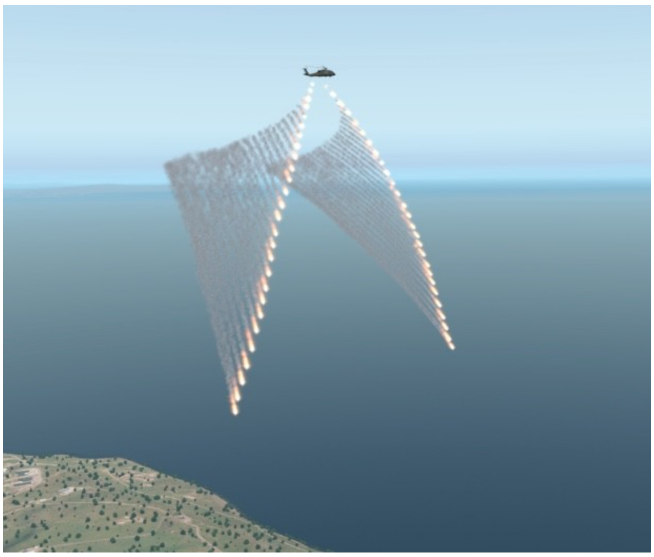
In addition use the GUN, RKT and MSL buttons on the EICAS MFD to arm/select installed/configured guns, rockets or missiles individually.



Use command “uh60m/weapons/cycle_weapons” to cycle through installed weapons (GUN, RKT, MSL).

To fire an armed/selected weapon use command: “uh60m/weapons/fire_armed”.

The guarded switch “JETTISON” will eject ALL remaining “Flares” with a single click:



MISSILES AND ROCKETS

Missiles and Rockets can be configured via the setup/config menu. The needed ESSS Pylon will be added/installed automatically.

The missile targets can be selected using the "camera target locking" in the TAC screen on the MFD.

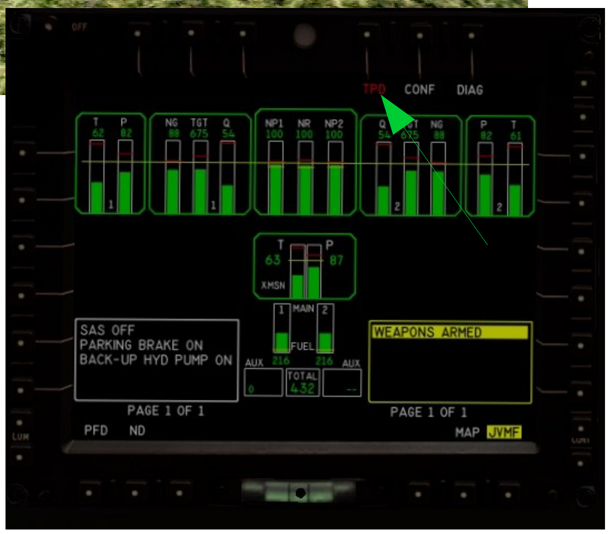
Rockets are unguided and just fly towards the aimed direction.

Alternatively the ESSS can be equipped with 0, 2 or 4 external fuel tanks. Mixing weapons and fuel tanks is not possible. Fuel tanks are prioritized and will remove missiles if selected.

Torpedo

The HM-60R (SeaHawk) is equipped with a Mark46 torpedo.

To launch it, select TPD in the EICAS display if Master-Arm switch has been engaged. Then the "fire-armed-weapon" command will kick it off. It will open a small parachute and falls slowly to the ground or water. If no target has been selected, it will follow the aircraft heading for about 15 seconds and at the end explodes. If a target has been selected in the CAM target display, it will swim/move to its location and explode at the end.

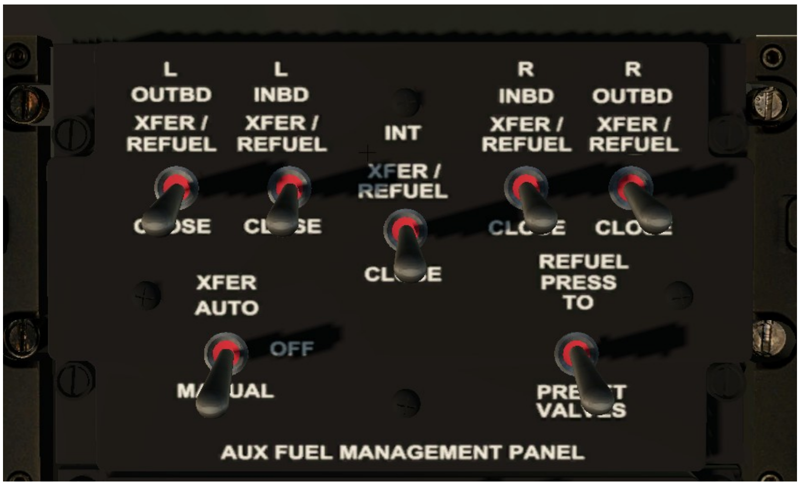


Aux Fuel Management Panel (AFMP)

The fuel management panel allows to move fuel from and to the AUX tanks.

There are 4 valves which control access to the 4 external tanks and one for the internal AUX tank.

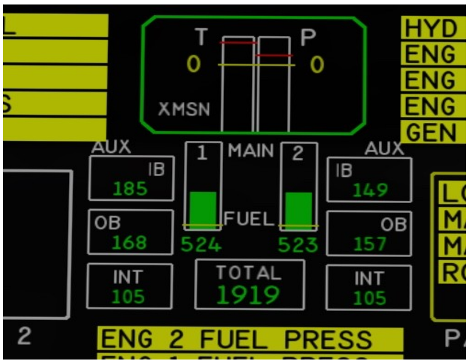
Each valve is controlled by an individual switch.



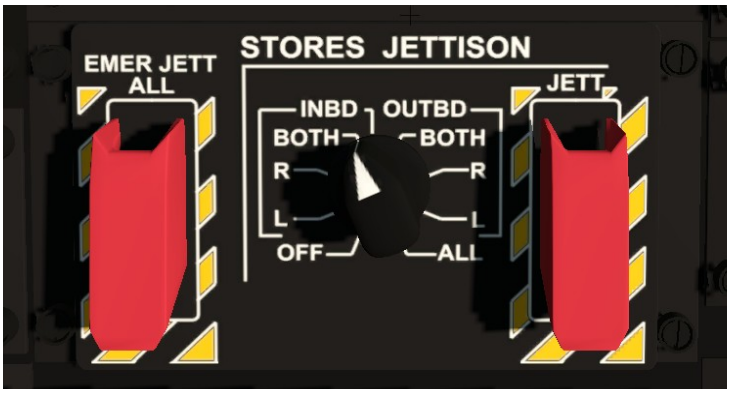
The XFER AUTO switch controls the valve and pump which transfers fuel <u>from</u> the selected external tanks <u>to</u> the internal main tanks. It can stay in AUTO during the whole flight and will keep the internal tanks filled, as long as fuel is in the selected external tanks. XFER MANUAL would fill the internal tanks from the pressure refueling.

The REFUEL switch controls the valve to fill the selected AUX tanks using “pressure refueling” from an external source like a fuel-truck. This works on ground only and the fuel truck and the fuel hose needs to be shown during refueling.

If external and/or internal AUX fuel tanks are configured, the fuel indicators in the MFD’s will also show the amount of fuel in the installed tanks.



The external fuel tanks can be jettisoned from the pedestal panel. Jettison will only work without WOW.



Animations


MAIN
CAMS
CONF
PREFS
BATTERIES & APU
CHECKLIST
COCKPIT DOORS
CARGO DOORS
GUNNER DOORS
ENGINE COWLINGS
AVIONICS COVER
HYDRAULICS COVER
TANK COVER
FUEL TRUCK
CONNECT GPU
1203
FLAMMABLE
NO SMOKING


ARMY
RIGHT STEPS


TAIL SERVICE LADDER


LEFT STEPS
UNITED ST

Videos

UH60M Videos playlist

https://www.youtube.com/watch?v=MuvI8dFNZUo&list=PLtspyUlCsgSZDCGbs1DjPPdm96kQ2mA7F

HowTo-clips playlist:

https://www.youtube.com/watch?v=AfriXZv96VE&list=PLtspyUlCsgSaIVPmxsf2exlWePyU4dzOE

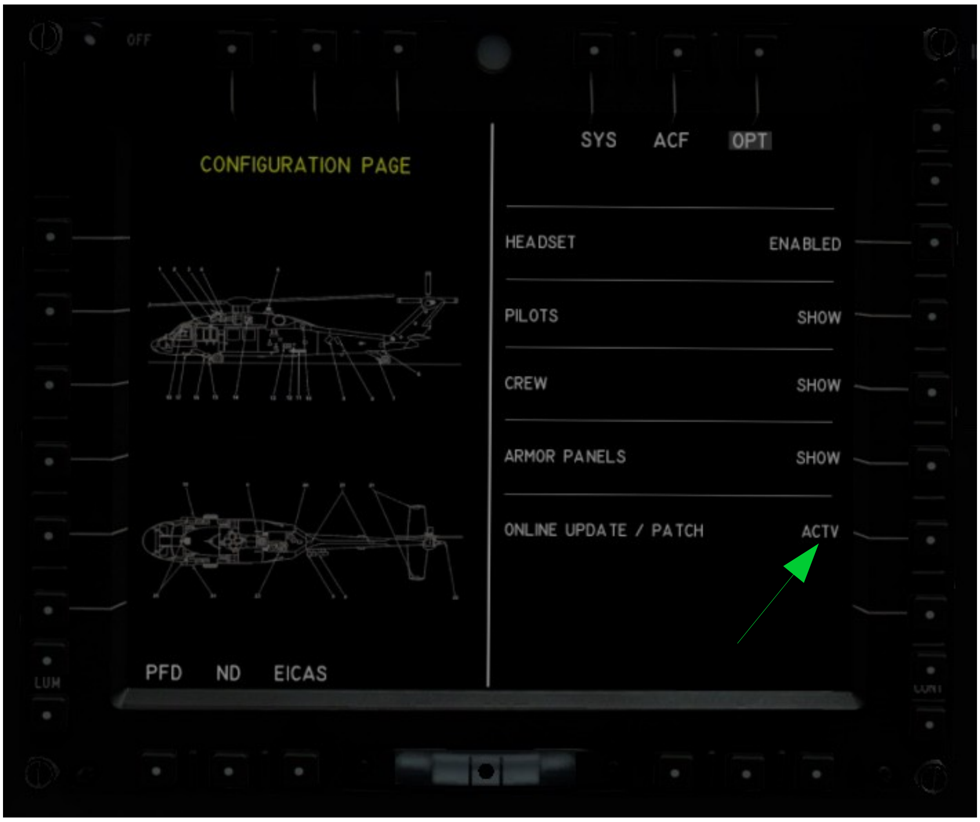
Make sure you subscribe to this YouTube channel to not miss new videos

Missions and Flightplans

see MISSIONS_HOWTO.TXT in “missions” folder

Updates

Online updates / patches can be enabled by activating them in the EICAS/CONF/OPT menu:



after enabling updates, you will see the “Update” menu:

Config Files

The Blackhawk maintains 2 config files.

“acf_config.ini” is the file which defines “global” aircraft settings which are independent from the used livery. The file resides in the aircraft base folder.

“opt_config.ini” is the file which defines options specific to a livery. Each livery has its own opt_config.ini.

Example acf_config.ini:

auto_hstab=1	Horizontal Stabilator auto mode active
auto_latch=1	Enable/disable auto throttle-idle latch
auto_vs=1	Enable/disable auto change of VS
auwarning=0	Aural Warnings
baro_hpa=0	Barometer Inch or HP
ctcycl=2	FTR cyclic mode 1=A , 2=B
dist_km=0	Distance KM or NM
headset=1	Headset
fps=1	FPS ON
fps_dz=0.06	FPS deadzone
ftr_dz=0.05	FTR deadzone
fuel1=324.84594726563 fuel2=324.84594726563	Last fuel amount
fuel_kg=1	Fuel KG or LBS
hw_throttle=0	HW throttle 0=n/a , 1=single, 2=dual
iglass=1	Inside Glass Reflections
patch=1	Plugin will check for updates
pilots=1	Show Pilots
sas1=1	Remember SAS1 switch setting
sas2=1	Remember SAS2 switch setting
temp_c=1	Temperature C or F
timezone=1	Timezone UTC or LOCAL
trim=1	Set TRIM ON
verbose=3	Logfile verbosity level
winch_ops=1	Disable internal winch code

Example opt_config.ini:

armorpnl=0	Show pilots armor panel
crew=1	Show crew
doors=1	Show cockpit doors
eoms=1	Show EOMS
esss=0	Show ESSS
exterior=0	Ext. Objects 1=SeaHawk 2=JayHawk 3=PaveHawk
flare=0	Show Flares/Chaffs
flir=1	Show FLIR cam
guns=1	Show GUNS
hoist=1	Show hoist winch
hrst=1	Show HRST ropes
intank=1	Install internal AUX tank
irjam=1	Show IR jammer
medevac=1	Show MedEvac interior
missiles=0	Show missiles
pax=0	Show pax (0 – 11)
probe=0	Show refuel probe
radar=0	Show nose radar 1=large 2=small
seats=1	Show Seats (0 – 3 rows)
tailnum=12345	Show tail number on dash
tailwl=1	0-1 tail dragger , 2 = Oleo (seahawk)
tanks=0	Install external AUX tanks (0 – 2 = 0 , 2 , 4)
variant=2	Door config 1-3 = Pavehawk, Seahawk, VIP
vh60=0	Marine-One config
vip=0	VIP interior config

Datarefs and Commands

Door commands

uh60m/doors/door_right_open	Cockpit doors
uh60m/doors/door_right_close	
uh60m/doors/door_right_toggle	
uh60m/doors/door_left_open	
uh60m/doors/door_left_close	
uh60m/doors/door_left_toggle	
uh60m/doors/cargo_right_open	Cargo doors
uh60m/doors/cargo_right_close	
uh60m/doors/cargo_right_toggle	
uh60m/doors/cargor_left_open	
uh60m/doors/cargo_left_close	
uh60m/doors/cargo_left_toggle	
uh60m/doors/gun_right_open	Gunner doors
uh60m/doors/gun_right_close	
uh60m/doors/gun_right_toggle	
uh60m/doors/gun_left_open	
uh60m/doors/gun_left_close	
uh60m/doors/gun_left_toggle	
uh60m/doors/avionics_open	Avionics bay door
uh60m/doors/avionics_close	
uh60m/doors/avionics_toggle	
uh60m/doors/hydraulics_open	Hydraulics bay door
uh60m/doors/hydraulics_close	
uh60m/doors/hydraulics_toggle	
uh60m/doors/engines_open	Engine cowlings
uh60m/doors/engines_close	
uh60m/doors/engines_toggle	
uh60m/doors/toggle_all	Toggle all doors
uh60m/doors/toggle_one	Select door by dataref
uh60m/doors/load_door_preset	dref: uh60m/doors/preset

Animation commands

uh60m/anims/bed_lft_upr_up	Move upper left bed up (MedEvac)
uh60m/anims/bed_lft_upr_dn	Move upper left bed down
uh60m/anims/bed_lft_lwr_up	Move lower left bed up
uh60m/anims/bed_lft_lwr_dn	Move lower left bed down
uh60m/anims/bed_rgt_upr_up	Move upper right bed up
uh60m/anims/bed_rgt_upr_dn	Move upper right bed down
uh60m/anims/bed_rgt_lwr_up	Move lower right bed up
uh60m/anims/bed_rgt_lwr_dn	Move lower right bed down

Light commands

uh60m/lights/ldg_lt_on	Landing light toggle
uh60m/lights/srch_lt_on	Search light toggle
uh60m/lights/srch_lt_up	Search light up
uh60m/lights/srch_lt_dn	Search light down
uh60m/lights/srch_lt_lft	Search light left
uh60m/lights/srch_lt_rgt	Search light right
uh60m/lights/dome_blue	Cargo dome light blue
uh60m/lights/dome_white	Cargo dome light white
uh60m/lights/light_dn	Dim cockpit lights
uh60m/lights/light_up	Brighten cockpit lights
uh60m/lights/cockpit_flood	Bright flood light toggle

System and Engines commands

uh60m/engn/start_engine0	Start engine 1
uh60m/engn/start_engine1	Start engine 2
uh60m/engn/idle_stop0	Release idle stop latch 1
uh60m/engn/idle_stop1	Release idle stop latch 2
uh60m/sys/ext_pwr_on	External Power ON
uh60m/sys/ext_pwr_test	External Power TEST/OFF
uh60m/sys/stby_inst_on	Standby Instrument ON
uh60m/sys/stby_inst_test	Standby Instrument TEST/OFF
uh60m/sys/batt1_on	Battery 1 ON
uh60m/sys/batt1_test	Battery 1 TEST/OFF
uh60m/sys/batt2_on	Battery 2 ON

uh60m/sys/batt2_test	Battery 2 TEST/OFF
uh60m/sys/apu_gen_on	APU Generator ON
uh60m/sys/apu_gen_test	APU Generator TEST/OFF
uh60m/sys/gen1_on	Generator 1 ON
uh60m/sys/gen1_test	Generator 1 TEST/OFF
uh60m/sys/gen2_on	Generator 2 ON
uh60m/sys/gen2_test	Generator 2 TEST/OFF
uh60m/sys/hover_hold	arm/disable hover hold
uh60m/sys/level_hold	arm/disable level hold
uh60m/sys/sas_boost	en/disable SAS boost
uh60m/sys/hstab_up	Move stabilator manually up
uh60m/sys/hstab_dn	Move stabilator manually down
uh60m/sys/clock_sel	Select clock/timer mode
uh60m/sys/clock_ctrl	start/stop/reset timer
uh60m/sys/ics_tx_nxt	Next TX device
uh60m/sys/ics_tx_prv	Prev TX device
uh60m/sys/ics_on	ICS comm and runup talk
uh60m/sys/wiper_up	Wiper up
uh60m/sys/wiper_dn	Wiper down
uh60m/sys/ftr	Force Trim Release
uh60m/sys/trim_reset	Reset pitch and roll trim
uh60m/sys/beep_fwd	Beep trip forward
uh60m/sys/beep_rev	Beep trim backward
uh60m/sys/beep_lft	Beep trim left
uh60m/sys/beep_rgt	Beep trim right
uh60m/caution_ack	Acknowledge caution warnings
uh60m/menu	Open setup menu

OPS commands

uh60m/ops/stow_rotor	Fold/unfold Rotor & Tail
uh60m/ops/tie_down	Tie-Down and Remove-Before-Flight
uh60m/ops/winch_prep	
uh60m/ops/winch_ena	
uh60m/ops/winch_conn	
uh60m/ops/winch_up	
uh60m/ops/winch_down	
uh60m/ops/rescuer_hook_only	

uh60m/ops/ipad_home	Ipad home button
uh60m/ops/ipad_toggle	Ipad show toggle
uh60m/ops/fuel_probe_out	Extend fuel probe
uh60m/ops/fuel_probe_in	Retract fuel probe
uh60m/ops/fuel_probe_toggle	Toggle fuel probe
uh60m/ops/rope_out	Show ropes
uh60m/ops/rope_in	Hide ropes
uh60m/ops/drop_dude	Drop seal
uh60m/ops/extract_seal	Place Seal at OSM pos and set FMS
uh60m/ops/iff_emerg	Sets transponder to 7700

Nav commands

uh60m/nav/nav_src_up	Select next RNAV source
uh60m/nav/nav_src_dn	Select previous RNAV source

Weapon commands

uh60m/weapons/master_arm	Master Arm switch
uh60m/weapons/set_master_arm	Master Arm for permanent switches
uh60m/weapons/fire_armed	Fire selected weapon
uh60m/weapons/fire_guns	Fire GUNS
uh60m/weapons/fire_rocket	Fire ROCKETS
uh60m/weapons/fire_missile	Fire MISSILES
uh60m/weapons/deploy_flares	Deploy FLARES
uh60m/weapons/deploy_chaff	Deploy CHAFF
uh60m/weapons/cycle_weapons	Cycle through weapons
uh60m/weapons/drop_torpedo	Launch torpedo

Camera commands

uh60m/cam/p1_select	Select Pilot camera
uh60m/cam/p2_select	Select CoPilot camera
uh60m/cam/c1_select	Select CrewChief camera
uh60m/cam/g1_select	Select ground crew 1 camera
uh60m/cam/g2_select	Select ground crew 2 camera
uh60m/cam/hoist_select	Select hoist camera
uh60m/cam/add	Add position to path
uh60m/cam/next	Go to next path position
uh60m/cam/reset	Go to first path position

uh60m/cam/clear	Clear path
uh60m/cam/save	Save paths to new.cam
uh60m/cam/update	Update current position
uh60m/cam/load	Load paths from default.cam
uh60m/cam/load_preflight	Load preflight path

MFD commands

uh60m/mfd/cam_power	FLIR Camera power
uh60m/mfd/cam_lock	FLIR Camera locked to target
uh60m/mfd/mfd1_pwr	Power button on MFD1 and MFD4
uh60m/mfd/mfd2_pwr	Power button on MFD2
uh60m/mfd/mfd3_pwr	Power button on MFD3
uh60m/mfd/mfd1_t1	MFD1 top row button 1
uh60m/mfd/mfd1_t2	MFD1 top row button 2 … 6
uh60m/mfd/mfd1_r1	MFD1 right row button 1
uh60m/mfd/mfd1_r2	MFD1 right row button 2 …. 6
uh60m/mfd/mfd1_b1	MFD1 bottom row button 1
uh60m/mfd/mfd1_b2	MFD1 bottom row button 2 … 6
uh60m/mfd/mfd1_l1	MFD1 left row button 1
uh60m/mfd/mfd1_l2	MFD1 left row button 2 … 6
uh60m/mfd/q1up	Queue 1 (yellow) msg box up
uh60m/mfd/q1dn	Queue 1 (yellow) msg box down
uh60m/mfd/q2up	Queue 2 (white) msg box up
uh60m/mfd/q2dn	Queue 2 (white) msg box down
uh60m/mfd/mfd_inc	Increment selected instrument
uh60m/mfd/mfd_dec	Decrement selected instrument
uh60m/mfd/mfd_inc2	Increment fast sel. Instr.
uh60m/mfd/mfd_dec2	Decrement fast sel. Instr.
uh60m/mfd/mfd_push	Push selected instrument
uh60m/mfd/mfd1_s1	MFD 1 select field 1
uh60m/mfd/mfd1_s2	MFD 1 select field 2
uh60m/mfd/mfd1_s3	MFD 1 select field 3
uh60m/mfd/mfd1_s4	MFD 1 select field 4
uh60m/mfd/mfd1_s5	MFD 1 select field 5
uh60m/mfd/mfd1_s6	MFD 1 select field 6

FDCP commands

uh60m/mfd/fdcp1d11	FDCP 1 Display 1 button 1
uh60m/mfd/fdcp1d12	FDCP 1 Display 1 button 2
uh60m/mfd/fdcp1d13	
uh60m/mfd/fdcp1d21	
uh60m/mfd/fdcp1d22	
uh60m/mfd/fdcp1d23	
uh60m/mfd/fdcp1d31	
uh60m/mfd/fdcp1d32	
uh60m/mfd/fdcp1d33	
uh60m/mfd/fdcp1d41	
uh60m/mfd/fdcp1d42	
uh60m/mfd/fdcp1d51	
uh60m/mfd/fdcp1d52	
uh60m/mfd/fdcp2d33	
uh60m/mfd/fdcp1_push1	Push rotary 1
uh60m/mfd/fdcp1_push2	Push rotary 2
uh60m/mfd/fdcp1_push3	Push rotary 3
uh60m/mfd/fdcp1_push4	Push rotary 4
uh60m/mfd/fdcp1_push5	Push rotary 5
uh60m/mfd/fdcp1_but_nxt	Select next button
uh60m/mfd/fdcp1_but_prv	Select previous button
uh60m/mfd/fdcp1_but_psh	Push selected button
uh60m/mfd/fdcp1_rot_nxt	Select next rotary
uh60m/mfd/fdcp1_rot_prv	Select previous rotary
uh60m/mfd/fdcp1_rot_psh	Push selected rotary
uh60m/mfd/fdcp1_rot_inc	Increment selected rotary
uh60m/mfd/fdcp1_rot_dec	Decrement selected rotary

CDU/FMS commands

uh60m/cdu/CDU1-popup	Open CDU1 popup
uh60m/cdu/CDU2-popup	Open CDU2 popup
uh60m/cdu/CDU1_L1	
uh60m/cdu/CDU1_L2	

uh60m/cdu/CDU1_L3	
uh60m/cdu/CDU1_L4	
uh60m/cdu/CDU1_L5	
uh60m/cdu/CDU1_R1	
uh60m/cdu/CDU1_R2	
uh60m/cdu/CDU1_R3	
uh60m/cdu/CDU1_R4	
uh60m/cdu/CDU1_R5	
uh60m/cdu/CDU1_CLR	
uh60m/cdu/CDU1_COM	
uh60m/cdu/CDU1_NAV	
uh60m/cdu/CDU1_FPN	
uh60m/cdu/CDU1_DAT	
uh60m/cdu/CDU1_INI	
uh60m/cdu/CDU1_XPDR	
uh60m/cdu/CDU1_ZRO	
uh60m/cdu/CDU1_ENT	
uh60m/cdu/CDU1_NXT	
uh60m/cdu/CDU1_PRV	
uh60m/cdu/CDU1_DIM	
uh60m/cdu/CDU1_BRT	
uh60m/cdu/CDU1_N0	
uh60m/cdu/CDU1_N1	
uh60m/cdu/CDU1_N2	
uh60m/cdu/CDU1_N3	
uh60m/cdu/CDU1_N4	
uh60m/cdu/CDU1_N5	
uh60m/cdu/CDU1_N6	
uh60m/cdu/CDU1_N7	
uh60m/cdu/CDU1_N8	
uh60m/cdu/CDU1_N9	
uh60m/cdu/CDU1_NMIN	
uh60m/cdu/CDU1_NDOT	
uh60m/cdu/CDU1_LFT	Cursor left
uh60m/cdu/CDU1_RGT	Cursor right
uh60m/cdu/CDU1_UP	Cursor up

uh60m/cdu/CDU1_DN	Cursor down
uh60m/cdu/com1_preset_up	Set COM1 to next preset
uh60m/cdu/com1_preset_down	Set COM1 to previous preset
uh60m/cdu/com2_preset_up	Set COM2 to next preset
uh60m/cdu/com2_preset_down	Set COM2 to previous preset

Mission commands

uh60m/mission/start	Start mission
uh60m/mission/stop	Stop/Hold mission
uh60m/mission/load	Load mission from file
uh60m/mission/download	Download mission from server
uh60m/mission/save	
uh60m/mission/reset	
uh60m/mission/clear	
uh60m/mission/add_wpt	Add current position to WPT list
uh60m/mission/skip	Skip current wpt or task
uh60m/mission/delete	
uh60m/mission/insert	
uh60m/mission/add	
uh60m/mission/sar_start_creep	Start a default creep-line search pattern
uh60m/mission/sar_start_square	Start a default square search pattern
uh60m/mission/sar_stop	Stop / hold the running search run
uh60m/mission/sar_cont	Continue the stopped search run
uh60m/mission/sar_clear	Stop and clear the generated pattern

Airfuel commands

uh60m/ops/c130_show	Toggle aerial refueling C130

Datarefs

uh60m/cdu/mission_name	Name of selected mission file name
uh60m/door/preset	Selected door preset
uh60m/conf/ipad	Show/hide iPad
uh60m/conf/probe	Refuel probe installed
uh60m/conf/probe_ext	Refuel probe extended
uh60m/conf/hoist	Hoist winch installed
uh60m/conf/homing	Homing antennas
uh60m/conf/hrst	Rope mount installed
uh60m/conf/doors	Cockpit doors installed
uh60m/conf/exterior	Add exterior objs. 1=sea 2=jay 3=pavehawk
uh60m/conf/medevac	Show MedEvac interior
uh60m/conf/seats	Cargo seats installed
uh60m/conf/seats2	Crew chief seat
uh60m/conf/pax	Passengers /Seals on seats
uh60m/conf/tanks	1 or 2 will install 2 or 4 aux tanks
uh60m/conf/intank	Install internal AUX tanks
uh60m/conf/guns	Guns installed
uh60m/conf/flare	Chaff and flares installed
uh60m/conf/missiles	Missiles and Rockets installed
uh60m/conf/esss	Tanks or weapons pylon installed
uh60m/conf/pilots	Show pilots
uh60m/conf/crew	Show crew , requires pilots
uh60m/conf/armorpnl	Armor panel in cockpit installed
uh60m/conf/flir	FLIR cam installed
uh60m/conf/irjam	IR jammer installed
uh60m/conf/eoms	Missile Sensors installed
uh60m/conf/radar	Radar installed 1=large 2=small
uh60m/conf/earplugs	Earplug / Headset
uh60m/conf/downwash	Show rotor downwash
uh60m/conf/iglass	Inside glass reflections
uh60m/conf/tailwl	Tailwheel 0=blackhawk , 1=seahawk
uh60m/conf/trim	TRIM ON
uh60m/conf/variant	Door config: 0=bh, 1=ph, 2=sh, 3=VIP
uh60m/conf/vibes	Cockpit vibrations

uh60m/conf/vip	VIP interior
uh60m/conf/vh60	Reconfig to VH-60N Marine-One
uh60m/sys/sw/hstab	HSTAB switch
uh60m/sys/sw/sas1	SAS1 switch
uh60m/sys/sw/sas2	SAS2 switch
uh60m/sys/sw/sas_boost	SAS boost switch
uh60m/sys/sw/tailwheel	Tailwheel lock switch
uh60m/sys/sw/egi1	EGI 1 switch
uh60m/sys/sw/egi2	EGI 2 switch
uh60m/sys/sw/ralt_test	Radio altimeter test
uh60m/sys/sw/hydr_bkup	Hydraulic backup switch
uh60m/sys/sw/air_src	Air source for starter
uh60m/sys/sw/stby_inst	Backup / Standby instrument
uh60m/sys/sw/zro_cdu	Zero CDU data
uh60m/sys/sw/srvo_off	Servo Tests
uh60m/sys/sw/ac_up_down	AntiCollision lower / both / upper
uh60m/sys/sw/ac_day_night	AntoCollision day / night
uh60m/sys/sw/nav_lights	Nav Light on / off
uh60m/sys/sw/nav_flash	Nav Light flash / steady
uh60m/sys/sw/lamp_test	Lamp test
uh60m/sys/sw/fire_test	Fire test
uh60m/sys/hover_mode	HSI Hover mode
uh60m/sys/hover_hold	Hover hold state
uh60m/sys/level_hold	Level hold state
uh60m/sys/fps_on	FPS / Heading-hold
uh60m/cam/selected	Selected camera
uh60m/ops/winch_ena	Winch enabled
uh60m/weapons/master_arm	Master weapons armed
uh60m/mfd/on	Array/4 – MFD on
uh60m/mfd/page	Array/4 – MFD page
uh60m/mfd/tgt_mode	
uh60m/mfd/hsi_mode	

uh60m/mfd/show_efis	
uh60m/mfd/show_wxr	**Show EFIS weather radar**
uh60m/sys/CB	**Array/100 – circuit breakers**

Functional Circuit Breakers

CB_hstab = 8 → Horizontal stabilator
CB_sas = 9 → SAS
CB_vhf1 = 12 → COM1 power
CB_tail_lock = 13 → Tailwheel lock
CB_master_wrn = 15 → Master Warn Panel
CB_egi2 = 16 → GPS2 power
CB_vhf2 = 17 → COM2 power
CB_hydr_pump = 24 → Hydraulic backup pump
CB_pri_fuel = 33 → APU primer boost
CB_egi1 = 34 → GPS1 power
CB_cefs1 = 38 → Fuel xfer pump via bus 1
CB_cefs2 = 45 → Fuel xfer pump via bus 2